

Introduction to Kubernetes

- Kubernetes word has its origin from Greek word “helmsman” which is someone who steers a ship, for example a containers ship
- Sometimes called k8s (there are 8 letters between “k” and “s”)
- Open-source container orchestrator which has all the capabilities/benefits of an orchestrator
- Used to automate the deployment, scaling, and management of containers
- It can run on-premises, public cloud and hybrid environments
- Kubernetes consists of a master-worker node architecture:
 - Master node(s): hosts the control plane which is responsible for maintaining the desired state of the cluster
 - Worker node(s): sometimes called data plane. Responsible for running the containers in pods. They report their status to the control plane
- A Kubernetes cluster contains nodes, deployments, pods, services and other Kubernetes objects
- About pods , deployments, nodes, and services:
 - Pod: small and simple unit that runs one or more containers. You can't run containers in Kubernetes without a pod. A pod has a unique IP in the cluster. Multiple containers inside a pod share storage and the same IP and can communicate with each other using localhost
 - Deployment: object to control the behavior of similar pods
 - Node: virtual or physical machine that runs pods
 - Service: object used to expose pods via a single IP address from a range called service CIDR. You may have multiple pods for the same application. You can use a service to load balance traffic between them and connect via a unique IP address: service's IP address